APPLIED PORTFOLIO RESEARCH

Constrained portfolio construction

Financial advisory internship · Fideuram - Intesa Sanpaolo Private Banking · June-September 2026

Finance, risk and quantitative research

I use Python to turn financial questions into structured analyses, test assumptions and explain what the evidence supports. My work connects portfolio construction, market simulation and research software with explicit risk controls.

The question

How can investment alternatives be compared under explicit allocation, concentration and risk constraints?

The work

A Python workflow for ETF and multi-asset screening, common-window risk and correlation analysis, and constrained allocation search. The saved optimization run enumerates 65,807 admissible candidate allocations on a 2.5% weight grid, with selection on the 2019-2024 design period and separate checks on 2025-2026 data.

Evidence and interpretation

Saved run manifests and methodology document the candidate count, allocation rules and separation between design and out-of-sample evaluation. Outputs support a transparent comparison of historical alternatives. Portfolio analytics is useful when the assumptions, constraints and selection process can be explained alongside the metrics.

Scope and limits

65,807 counts candidate allocations, not managed portfolios or client outcomes. Operational trading costs are excluded, the holdout is short, and market proxies are distinct from product returns. Historical comparisons do not predict future returns or guarantee drawdown limits. Internal files and holdings are excluded.

Code and reproducibility

Independent synthetic demo: 11 tests and CPU example passed on Python 3.12/Windows. Demonstrates constraints, design-only selection and risk accounting; original internship materials are excluded.

github.com/subbace/portfolio-construction-demo

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MASTER'S THESIS

Market regimes and robust hedging

Quantitative Finance and Risk Management · Ca' Foscari University of Venice · 2026

The question

Do neural hedging policies remain robust when market conditions differ from their development environment?

The work

Agent-based market simulation, neural hedging in PyTorch and comparison with a same-window Black-Scholes benchmark under transaction costs. The research includes fresh out-of-distribution stress scenarios and a causal transfer study on 139 historical SPY windows.

Evidence and interpretation

The recorded full studies include 100 fresh stress paths across ten seeds per neural family, and an evaluation-only closure on 139 historical SPY windows. The SPY study uses synthetic options on observed underlying-price paths, not historical traded-option quotes. Development-sample gains did not establish robust aggregate superiority. Black-Scholes remains the recorded winner on the historical SPY sample; negative findings are retained.

Scope and limits

The historical SPY sample is already inspected and cannot be reused as an untouched test or for model selection. Simulated hedging results do not establish investment returns or execution feasibility.

Code and reproducibility

Reduced synthetic thesis experiment: 8 tests, CPU run and numerical-reference comparison passed on Python 3.12/Windows. Preserved causal components; no neural weights, full-study replication or superiority claim.

github.com/subbace/abm-liquidity-hedging-demo

Background

Quantitative Finance and Risk Management · Ca' Foscari University of Venice · 2024-2026
Bachelor's degree in Digital Management · Ca' Foscari University of Venice / H-FARM College · 2020-2023

Italian native · English C1 · Python, pandas, NumPy, PyTorch and Excel

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CONTINUING RESEARCH

Real-options research infrastructure

Independent continuation of the thesis · active development · November 2026

The question

What infrastructure is needed to test a hedge using option-market data, explicit execution assumptions and a recoverable financial account?

The work

An event-driven Python laboratory connecting market-data provenance, causal observations, conservative liquidity scenarios, option and futures accounting, and transactional SQLite recovery. Recurrent policy prototypes connect observations, memory and financial state.

Evidence and interpretation

The frozen offline RO-01 foundation records 59 passing tests and twelve engineering gates. Later synthetic pilots verify complete option/futures round trips and independent cash reconciliation. A separate continuity exercise preserves state across 130 observations and 17 recovery segments. This work makes research decisions and account changes inspectable, including partial fills, uncertain acknowledgements, invalid data and interrupted execution.

Scope and limits

The current recurrent actors are untrained. The long-horizon exercise passed its functional checks but failed its CPU budget; it does not close a full financial horizon. Qualified market data, complete learners and economic validation remain open. No live trading or demonstrated profitability.

Code and reproducibility

Offline RO-01 subset: 59 original tests, 5 demo tests and the example passed on Python 3.12/Windows. No live data, recent neural components or economic validation.

github.com/subbace/real-options-infrastructure-demo

SQL and Business Intelligence: course-based knowledge, self-assessed at intermediate level.

Research and coding workflows use AI assistance, with documented checks and human responsibility for the final claims. The three entries describe separate workstreams; the third continues the thesis rather than representing an unrelated study.

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